

The Kronecker character χ_{-4} as the joint source of the lemniscatic identity algebra:

a unified treatment of $G^* = \Gamma(1/4)/\Gamma(3/4)$ and $G_G = 1/\text{AGM}(1, \sqrt{2})$

Abstract

The lemniscatic CM point $\tau = i$ generates a single transcendental constant whose value has two equally-natural normalisations: the *ratio of the Euler reflection at $z = 1/4$* , $G^* := \Gamma(1/4)/\Gamma(3/4) \approx 2.95867\dots$, and *Gauss's constant*, $G_G := 1/\text{AGM}(1, \sqrt{2}) \approx 0.83463\dots$. They are linearly related by $G^* = 2\sqrt{\pi} G_G$, and either may be chosen as the principal variable. We show that the choice is not arbitrary: every identity in this neighbourhood has a *natural form* with integer or rational coefficients that picks out exactly one of $\{G^*, G_G\}$. Specifically, identities arising from the *algebraic* structure (Euler reflection, Beta integrals at quarter arguments, the master quadratic with integer coefficient $16 = |\text{Aut}(E)|^2$ for the lemniscatic curve $E : y^2 = x^3 - x$) are natural in G^* ; identities arising from the *analytic* structure (modular forms at $\tau = i$, the body-centred-cubic Watson integral, AGM iteration, the Dedekind eta function) are natural in G_G .

We present this dichotomy as the organising principle of the paper. We catalogue over fifty identities in their natural form, including the clean evaluations

$$\begin{aligned} E_4(i) &= 3 G_G^4, & \Delta(i) &= G_G^{12}/64, & \eta(i) &= G_G^{1/2}/2^{1/4}, \\ W_{\text{BCC}}^{(3)} &= 2 G_G^2 = 2/\text{AGM}(1, \sqrt{2})^2, & W_{\text{BCC}}^{(D)} &= {}_D F_{D-1}(\tfrac{1}{2}, \dots; 1, \dots; 1), \end{aligned}$$

the reflection-form identities $\Gamma(1/4)^2 = \pi\sqrt{2} G^*$ and $\Gamma(1/4)^4 = 2\pi^2 G^{*2}$, and the master quadratic $x^2 - 16 G^{*2} x + 16 G^{*3} = 0$ whose integer coefficient 16 is the squared automorphism count of the lemniscatic curve.

Two structural results emerge: **(I)** the value algebra of quasi-modular forms at $\tau = i$ is the polynomial ring $\mathbb{Q}[\pi^{-1}, G_G^4]$ in two transcendently independent generators under Chudnovsky's theorem, with π^{-1} recording the quasi-modular anomaly and G_G^4 recording the modular weight; **(II)** the $m \mid k$ vanishing principle, stating that at a CM point with $|\text{Aut}(E)| = m$ a modular form f of weight k satisfies $f(\tau_{\text{CM}}) = 0$ unless $m \mid k$, holds in the only two non-trivial cases over $\mathbb{Q}$ (lemniscatic with $m = 4$, equianharmonic with $m = 6$). Every identity is traced to its earliest source; the systematic dual-constant presentation appears to be new.

A companion note [13] draws the consequences of the χ_{-4} -unification developed here for the Foundational Ternary Dynamics framework; the present paper does not depend on or address that framework.

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1 Introduction

The Gauss–Euler reflection formula

$$\Gamma(z) \Gamma(1-z) = \frac{\pi}{\sin \pi z}, \quad z \notin \mathbb{Z}, \quad (1)$$

relates two evaluations of the Gamma function symmetric about $z = 1/2$. At $z = 1/4$ the product

$$\Gamma(1/4) \Gamma(3/4) = \pi\sqrt{2}$$

is the unique linear combination of $\Gamma(1/4)$ and $\Gamma(3/4)$ that reduces to a closed form in π . The other natural combination is the ratio,

$$G^* := \frac{\Gamma(1/4)}{\Gamma(3/4)} = 2.95867511918863889231\dots \quad (2)$$

which yields a new transcendental.

Independently of (1), the arithmetic–geometric mean of Gauss provides a second natural lemniscatic CM constant:

$$G_G := \frac{1}{\text{AGM}(1, \sqrt{2})} = 0.83462684167407318628\dots \quad (3)$$

known classically as *Gauss’s constant*. A brief history clarifies its standing: Gauss recorded the central observation $\text{AGM}(1, \sqrt{2}) \cdot \varpi / \pi = 1$ (equivalent to $\text{AGM}(1, \sqrt{2}) = \pi / \varpi$) in his diary on 30 May 1799 [14, Werke III, p. 370], calling the discovery a result that “*will surely open an entirely new field of analysis*.” He kept the result essentially unpublished during his lifetime; the AGM theory appeared posthumously in his *Werke*, with most details remaining in his *Nachlass* until Geppert’s 1927 edition. Brent [2] and Salamin [3] independently rediscovered the AGM-based algorithm for π in 1976, recognising the connection to Gauss’s diary. The constant $G_G = 1/\text{AGM}(1, \sqrt{2})$ now bears Gauss’s name in standard references [4, 5]. The lemniscate constant $\varpi = \Gamma(1/4)^2 / (2\sqrt{2}\pi) \approx 2.62206\dots$ equals πG_G , and connects (2) and (3) by the chain of identities (proved in §2):

$$G^* = 2\sqrt{\pi} G_G, \quad \varpi = \pi G_G = \frac{G^* \sqrt{\pi}}{2}. \quad (4)$$

Either G^* or G_G may be chosen as the principal variable in this neighbourhood, but the choice has a clear structural fingerprint, and that fingerprint is the central observation of the present paper: *Observation* (Dichotomy of the natural form). Identities arising from the *algebraic* structure of the lemniscatic CM point—reflection ratios, Beta integrals at quarter arguments, the master quadratic with integer coefficient $16 = |\text{Aut}(E)|^2$, the family $R_n = \Gamma(1/n)/\Gamma((n-1)/n)$ —have rational coefficients in G^* . Identities arising from the *analytic* structure—modular forms at $\tau = i$, the body-centred-cubic lattice Green’s function, AGM iteration, the Dedekind eta function—have rational coefficients in G_G . Each identity has a natural form in exactly one of the two normalisations; the other introduces transcendental factors of π .

This dichotomy is structurally robust: $E_4(i) = 3G_G^4$ is integer-coefficient clean, while $E_4(i) = 3G^{*4}/(16\pi^2)$ requires a π^2 denominator. The master quadratic $x^2 - 16G^{*2}x + 16G^{*3} = 0$ has integer coefficients in G^* because 16 is the squared automorphism count of $E : y^2 = x^3 - x$ (an algebraic invariant of the lemniscatic curve); the same polynomial written in G_G is $x^2 - 64\pi G_G^2 x + 128\pi^{3/2} G_G^3 = 0$, which obliterates the integer structure with fractional π -powers.

This paper presents a systematic survey of identities at $\tau = i$ under this dual-constant discipline. We catalogue over fifty identities, each in its natural form, organised by the algebraic–analytic dichotomy. The component identities span the literature from Gauss (1799), Euler (1771), Watson (1939), Joyce (1971–72), Chowla–Selberg (1967), Borwein–Borwein (1987), Kontsevich–Zagier (2001), and Guttman (2010); we trace each to its earliest source. The synthetic contribution is the dual-constant framing as an organising principle, which appears not to have been stated explicitly elsewhere.

Structure of the paper. §2 establishes the bridge identity (4). §3–8 treat the algebraic side: reflection identities (§3), the three equivalent definitions of G^* (§4), the lemniscatic elliptic curve (§5), the master quadratic (§6), Beta integrals (§7), and the family R_n (§8). §9–12 treat the analytic side: the Watson lattice identity (§9), theta functions at $\tau = i$ (§10), the Dedekind eta function (§11), and modular forms at $\tau = i$ (§12, including the quasi-modular extension giving structural result (I) of the abstract). §13 addresses the Kontsevich–Zagier period status; §14 records transcendence and algebraic-independence status. §15 consolidates the dichotomy in a side-by-side compendium. §16 develops the equianharmonic dichotomy at $\tau = \rho$, yielding structural result (II) of the abstract. §17 presents the unifying χ_{-4} structure underlying both dichotomies, including the four-level projection theorem, three negative tests on the polynomial-assembly question, the asymptotic-regime analysis, the symmetric-algebra conjecture, and the ontological zero-point identification $\chi_{-4}(n) = \text{Im}(i^n)$. §18 states two open problems (KZ-strict status, cubic-AGM for G_ρ); §19 states two algebraic-independence conjectures supported by high-precision PSLQ evidence ($W_{\text{BCC}}^{(4)}$, Catalan’s constant).

2 The bridge $G^* = 2\sqrt{\pi} G_{\text{G}}$

Theorem 2.1. $G^* = 2\sqrt{\pi} G_{\text{G}}$.

Proof. By the Gauss–Legendre identity [1, Thm. 1.1], $K(k) = \pi/[2 \text{AGM}(1, \sqrt{1-k^2})]$. At $k = 1/\sqrt{2}$ (the self-complementary modulus), $\sqrt{1-k^2} = 1/\sqrt{2}$, so

$$K(1/\sqrt{2}) = \frac{\pi}{2 \text{AGM}(1, 1/\sqrt{2})} = \frac{\pi}{\sqrt{2} \text{AGM}(1, \sqrt{2})} = \frac{\pi G_{\text{G}}}{\sqrt{2}},$$

where we used the homogeneity $\text{AGM}(\lambda a, \lambda b) = \lambda \text{AGM}(a, b)$ with $\lambda = 1/\sqrt{2}$. The classical value $K(1/\sqrt{2}) = \Gamma(1/4)^2/(4\sqrt{\pi})$ [1, Eq. 2.5.4] gives

$$G_{\text{G}} = \frac{\sqrt{2} K(1/\sqrt{2})}{\pi} = \frac{\sqrt{2} \Gamma(1/4)^2}{4\pi^{3/2}}.$$

By the reflection identity (Lemma 3.1 below), $\Gamma(1/4)^2 = \pi\sqrt{2} G^*$, so

$$G_{\text{G}} = \frac{\sqrt{2} \cdot \pi\sqrt{2} G^*}{4\pi^{3/2}} = \frac{G^*}{2\sqrt{\pi}}. \quad \square$$

Corollary 2.2 (Conversion table). *The following identities convert between G^* , G_{G} , and the lemniscate constant ϖ :*

$G^* = 2\sqrt{\pi} G_{\text{G}}$	$G_{\text{G}} = G^*/(2\sqrt{\pi})$
$\varpi = \pi G_{\text{G}} = G^* \sqrt{\pi}/2$	$G_{\text{G}} = \varpi/\pi$
$G^{*2k} = (4\pi)^k G_{\text{G}}^{2k}$	$G_{\text{G}}^{2k} = G^{*2k}/(4\pi)^k$
$G^{*2k+1} = 2^{2k+1} \pi^{k+1/2} G_{\text{G}}^{2k+1}$	$G_{\text{G}}^{2k+1} = G^{*2k+1}/(2^{2k+1} \pi^{k+1/2})$

Throughout the paper, we adopt the discipline: *express each identity in the convention in which its coefficients are rational*, with the bridge identity supplied where conversion is needed. This is the operational content of Observation 1.

3 Reflection identities: the algebraic core in G^*

Lemma 3.1 (Reflection at $z = 1/4$).

$$\Gamma(1/4)^2 = \pi\sqrt{2} G^*, \quad (5)$$

$$\Gamma(3/4)^2 = \pi\sqrt{2}/G^*, \quad (6)$$

$$\Gamma(1/4)^4 = 2\pi^2 G^{*2}, \quad (7)$$

$$\Gamma(1/4)^8 = 4\pi^4 G^{*4}. \quad (8)$$

Proof. Euler's reflection (1) at $z = 1/4$: $\Gamma(1/4)\Gamma(3/4) = \pi/\sin(\pi/4) = \pi\sqrt{2}$. Multiply by G^* for (5); divide for (6); square for (7) and (8). $\square$

Remark. These four identities exhibit the cleanness of G^* for the reflection-channel structure: rational coefficients in G^* (no π inside the coefficient), with π appearing only as the external factor from Euler reflection. The corresponding identities in G_G have the form $\Gamma(1/4)^2 = (\pi\sqrt{2})(2\sqrt{\pi})G_G = 2\sqrt{2}\pi^{3/2}G_G$, which introduces $\pi^{3/2}$ — fractional powers.

Corollary 3.2. *The two complementary identities are:*

$$\Gamma(1/4)\Gamma(3/4) = \pi\sqrt{2} \quad (\text{product channel, yields } \pi), \quad (9)$$

$$\Gamma(1/4)/\Gamma(3/4) = G^* \quad (\text{ratio channel, yields the new transcendental}). \quad (10)$$

The pair $(\Gamma(1/4), \Gamma(3/4))$ is uniquely determined by these two quantities:

$$\Gamma(1/4) = \sqrt{\pi\sqrt{2} G^*}, \quad \Gamma(3/4) = \sqrt{\pi\sqrt{2}/G^*}.$$

4 Three equivalent definitions of G^*

Theorem 4.1 (Equivalent definitions). *The following are equal:*

- (i) $G^* = \Gamma(1/4)/\Gamma(3/4)$ (*Gamma-ratio form*).
- (ii) $G^* = \sqrt{2\pi}/\text{AGM}(1, 1/\sqrt{2})$ (*AGM form*).
- (iii) $G^* = \frac{1}{\sqrt{2\pi}} \int_0^1 \frac{dt}{[t(1-t)]^{3/4}}$ (*Beta-integral form*).

Proof. (i) = (iii): The Beta integral gives $B(1/4, 1/4) = \Gamma(1/4)^2/\sqrt{\pi}$. By (5), $\Gamma(1/4)^2 = \pi\sqrt{2} G^*$, so $B(1/4, 1/4) = \sqrt{2\pi} G^*$, equivalent to (iii).

(i) = (ii): By homogeneity $\text{AGM}(\lambda a, \lambda b) = \lambda \text{AGM}(a, b)$, $\text{AGM}(1, 1/\sqrt{2}) = \text{AGM}(\sqrt{2}, 1)/\sqrt{2} = \text{AGM}(1, \sqrt{2})/\sqrt{2}$. By Theorem 2.1, $\text{AGM}(1, \sqrt{2}) = 1/G_G = 2\sqrt{\pi}/G^*$, so $\text{AGM}(1, 1/\sqrt{2}) = \sqrt{2\pi}/G^*$, which gives (ii). $\square$

Remark. Definition (i) is the algebraic-reflection form. Definition (ii) gives the bridge to G_G : rewriting (ii), $G^* = \sqrt{2\pi} \cdot \sqrt{2} \cdot G_G = 2\sqrt{\pi} G_G$, which recovers (4). Definition (iii) is the period-integral form relevant for Kontsevich–Zagier (§13).

5 The lemniscatic elliptic curve

Definition 5.1. The lemniscatic elliptic curve over $\mathbb{Q}$ is $E : y^2 = x^3 - x$.

Lemma 5.2. E has complex multiplication by $\mathbb{Z}[i]$, j -invariant 1728, discriminant 64, and $|\text{Aut}_{\mathbb{C}}(E)| = 4$, with the automorphism group generated by $(x, y) \mapsto (-x, iy)$.

Theorem 5.3 (Real period). *The real period of E admits the three equivalent expressions:*

$$\omega_E = 2 \int_{-1}^0 \frac{dx}{\sqrt{x(x-1)(x+1)}} \quad (11)$$

$$= \frac{\Gamma(1/4)^2}{\sqrt{2\pi}} \quad (\text{Gamma form}) \quad (12)$$

$$= G^* \sqrt{\pi} \quad (\text{reflection form, in } G^*) \quad (13)$$

$$= \pi \sqrt{\pi} \cdot 2G_G \quad (\text{AGM form, in } G_G; \text{ introduces } \pi^{3/2}). \quad (14)$$

The first three forms have rational/algebraic coefficients; the fourth exhibits the $\pi^{3/2}$ factor that characterises this quantity as algebraic by origin but not cleanly expressible in G_G .

Proof. The integral evaluation $\omega_E = \Gamma(1/4)^2/\sqrt{2\pi}$ is classical; see [1, §2.5]. The G^* -form follows from $\Gamma(1/4)^2 = \pi\sqrt{2} G^*$ (Lemma 7.1). The G_G -form follows from the bridge identity $G^* = 2\sqrt{\pi} G_G$: $\omega_E = G^* \sqrt{\pi} = (2\sqrt{\pi} G_G) \sqrt{\pi} = 2\pi G_G$. $\square$

Observation. Theorem 5.3 reveals that $\omega_E = 2\pi G_G$ has clean integer coefficient in G_G (the period is literally 2π times Gauss's constant) but also has clean form $G^* \sqrt{\pi}$. This is one of the borderline identities where both conventions are reasonable. The lemniscatic period sits exactly at the algebraic-analytic boundary: it is the period of an algebraic curve (algebraic origin, hence G^* representation $\omega_E = G^* \sqrt{\pi}$ is natural) but also the quadratic-mean of two AGM iterations (analytic origin, hence $\omega_E = 2\pi G_G$). Both forms are integer-coefficient.

6 The master quadratic: the algebraic centerpiece

Consider the polynomial

$$P_{G^*}(x) := x^2 - 16 G^{*2} x + 16 G^{*3} \quad (15)$$

with integer coefficient 16.

Theorem 6.1 (Master quadratic structure). *The polynomial $P_{G^*}(x)$ has positive discriminant $256 G^{*4} - 64 G^{*3} = 64 G^{*3}(4G^* - 1) > 0$ and roots*

$$x_{\pm} = 8 G^{*2} \pm 4 G^{*3/2} \sqrt{4G^* - 1}.$$

Numerically:

$$x_+ = 137.03617145815548388 \dots, \quad (16)$$

$$x_- = 3.02396391633902100 \dots \quad (17)$$

Vieta's identities give

$$\frac{1}{x_+} + \frac{1}{x_-} = \frac{1}{G^*}, \quad x_+ \cdot x_- = 16 G^{*3}. \quad (18)$$

Proof. Direct computation via the quadratic formula and Vieta's identities. $\square$

Observation (Integer-coefficient signature). The polynomial P_{G^*} has integer coefficients in G^* . The coefficient $16 = |\text{Aut}_{\mathbb{C}}(E)|^2$ is the squared automorphism count of the lemniscatic curve. In G_G -coordinates, the same polynomial becomes

$$x^2 - 64\pi G_G^2 x + 128\pi^{3/2} G_G^3,$$

with explicit π and $\pi^{3/2}$ in the coefficients. The integer 16 disappears; the algebraic identification with $|\text{Aut}(E)|^2$ is obscured. This is the cleanest illustration of Observation 1: the master quadratic is intrinsically a G^* object.

Remark (Numerical content of the roots). The roots take the numerical values $x_+ \approx 137.036$ and $x_- \approx 3.024$. Their identification with physical observables and the resulting math–physics bridge lie outside the mathematical scope of the present paper; the companion note [13] treats those questions. The transcendence and algebraic-independence implications of the roots are addressed in §14.

7 Beta integrals at quarter arguments

Lemma 7.1. *The Beta function at $(1/4, 1/4)$ admits the closed forms:*

$$B(1/4, 1/4) = \int_0^1 [t(1-t)]^{-3/4} dt \quad (19)$$

$$= \frac{\Gamma(1/4)^2}{\Gamma(1/2)} = \frac{\Gamma(1/4)^2}{\sqrt{\pi}} \quad (20)$$

$$= \sqrt{2\pi} G^* \quad (\text{in } G^*, \text{ clean}) \quad (21)$$

$$= 2\sqrt{2} \pi G_G \quad (\text{in } G_G, \text{ integer-coefficient}). \quad (22)$$

Proof. The Beta-integral evaluation is classical: $B(p, q) = \Gamma(p)\Gamma(q)/\Gamma(p+q)$. At $p = q = 1/4$, $\Gamma(p+q) = \Gamma(1/2) = \sqrt{\pi}$. The G^* -form follows from (5). The G_G -form follows from (4): $\sqrt{2\pi} G^* = \sqrt{2\pi} \cdot 2\sqrt{\pi} G_G = 2\sqrt{2\pi^2} G_G = 2\sqrt{2} \pi G_G$. $\square$

Remark. This is another borderline identity where both forms are clean: G^* gives $\sqrt{2\pi}$ as coefficient (irrational but classical), G_G gives 4π (integer multiple of π). Both are rational-coefficient identities in their respective constants. The Beta integral, like the lemniscatic period (5.3), sits at the boundary.

8 The family R_n and the distinguishedness of $R_4 = G^*$

Definition 8.1. For integer $n \geq 2$, $R_n := \Gamma(1/n)/\Gamma((n-1)/n)$.

Proposition 8.2. *The family $\{R_n\}_{n \geq 2}$ satisfies:*

- (i) $R_n > 0$ for all $n \geq 2$;
- (ii) $R_n = \Gamma(1/n)^2 \sin(\pi/n)/\pi$ (by Euler reflection);
- (iii) $R_2 = 1$ (trivially) and $R_n = n - 2\gamma + O(1/n)$ as $n \rightarrow \infty$, where γ is the Euler–Mascheroni constant.

The first six values:

n	R_n	associated CM-curve interpretation
2	1.0000000000	trivial
3	1.9783642596...	equianharmonic $E : y^2 = x^3 - 1$, $\text{End}(E) = \mathbb{Z}[\rho]$, $ \text{Aut} = 6$
4	2.9586751192... (G^*)	lemniscatic $E : y^2 = x^3 - x$, $\text{End}(E) = \mathbb{Z}[i]$, $ \text{Aut} = 4$
5	3.9432456137...	no CM curve with this argument structure
6	4.9312366764...	equianharmonic via twist $y^2 = x^3 + 1$
8	6.9261841305...	no CM curve

Theorem 8.3 (Algebraic distinguishedness of R_4). *Among the R_n family with $n \in \{2, 3, 4, 5, 6, 8, 12\}$, the value $R_4 = G^*$ is the unique value such that R_n is the gamma-ratio of a class-number-one CM elliptic curve over $\mathbb{Q}$ with $|\text{Aut}_{\mathbb{C}}(E)|^2 = 16$. The equianharmonic curve E_ρ at R_3 (and its twist at R_6) has $|\text{Aut}_{\mathbb{C}}(E_\rho)|^2 = 36$; no other R_n in the listed range corresponds to a CM elliptic curve with extra automorphisms.*

Proof. Direct enumeration. The class-number-one imaginary-quadratic fields with unit group strictly larger than $\{\pm 1\}$ are exactly $\mathbb{Q}(i)$ (units of order 4) and $\mathbb{Q}(\rho)$ (units of order 6). These yield the lemniscatic curve $y^2 = x^3 - x$ over $\mathbb{Q}$, with $R_4 = \Gamma(1/4)/\Gamma(3/4)$ and $|\text{Aut}|^2 = 16$, and the equianharmonic curve $y^2 = x^3 - 1$ (with twist $y^2 = x^3 + 1$), with $R_3 = \Gamma(1/3)/\Gamma(2/3)$ (resp. R_6) and $|\text{Aut}|^2 = 36$. For $n \in \{2, 5, 8, 12\}$, no CM elliptic curve over $\mathbb{Q}$ has R_n as its gamma-ratio. $\square$

Observation (Numerical content of the roots at R_4 and R_3). Numerical evaluation of $x_{\pm}(R_n) = (16R_n^2 \pm \sqrt{(16R_n^2)^2 - 64R_n^3})/2$ for the CM-relevant values R_3, R_4, R_6 yields:

n	R_n	$x_+(R_n)$	$x_-(R_n)$
3	1.97836	138.894 ...	2.007 ...
4	2.95868 (G^*)	137.036 ...	3.024 ...
6	4.93124	384.078 ...	4.995 ...

(In rows 3, 6 the polynomial coefficient is $16R_n^2/16R_n^3$ as in the lemniscatic case; the corresponding equianharmonic master quadratic $y^2 - 36R_n^2y + 36R_n^3$ would have $|\text{Aut}|^2 = 36$ as its integer prefactor.) Their interpretation as physical observables is addressed in the companion note [13].

Proposition 8.4 (Family-wide master-quadratic root asymptotic, with full expansion). *For the master-quadratic family $x^2 - 16R^2x + 16R^3 = 0$ with $R > 1/4$, the smaller root admits the absolutely convergent expansion*

$$x_-(R) = \sum_{n=1}^{\infty} d_n R^{2-n}, \quad d_n = \frac{\binom{2n}{n}}{(2n-1)2^{4n-3}},$$

giving explicit leading terms

$$x_-(R) = R + \frac{1}{16} + \frac{1}{128R} + \frac{5}{4096R^2} + \frac{7}{32768R^3} + O(R^{-4}).$$

The larger root satisfies $x_+(R) = 16R^2 - x_-(R)$ by Vieta, hence $x_+(R) = 16R^2 - R - 1/16 - 1/(128R) - O(R^{-2})$.

Proof. Write the quadratic formula in the form $x_-(R) = 8R^2(1 - \sqrt{1-y})$ with $y = 1/(4R) \in (0, 1)$. The Newton binomial expansion gives $1 - \sqrt{1-y} = \sum_{n \geq 1} c'_n y^n$ with $c'_n = \binom{2n}{n}/((2n-1)4^n)$ for $n \geq 1$, so

$$x_-(R) = 8R^2 \sum_{n \geq 1} \frac{c'_n}{(4R)^n} = \sum_{n \geq 1} \frac{8c'_n}{4^n} R^{2-n} = \sum_{n \geq 1} d_n R^{2-n},$$

where $d_n := 8c'_n/4^n = \binom{2n}{n}/((2n-1)2^{4n-3})$. Direct evaluation: $d_1 = 1$, $d_2 = 1/16$, $d_3 = 1/128$, $d_4 = 5/4096$, $d_5 = 7/32768$, recovering the leading R and the four subsequent correction terms. Convergence for $4R > 1$ is by ratio test: $|d_{n+1}R^{-1}/d_n| \rightarrow 1/(4R) < 1$. $\square$

Remark (Numerical verification across R_n). The difference $x_-(R_n) - R_n$ approaches $1/16 = 0.0625$ from above:

n	R_n	$x_-(R_n)$	$x_-(R_n) - R_n$
3	1.97836	2.04516	0.06680
4	2.95868 (G^*)	3.02396	0.06528
5	3.94325	4.00781	0.06456
6	4.93124	4.99537	0.06413
8	6.91408	6.97773	0.06366
12	10.89429	10.95752	0.06323
∞	$(n - 2\gamma)$	$(\rightarrow R_n + 1/16)$	$1/16 = 0.06250$

The constant excess $x_-(R_n) - R_n \rightarrow 1/16$ as $n \rightarrow \infty$ is a structural feature of the family $x^2 - 16R^2x + 16R^3 = 0$ and is independent of any physical interpretation of the roots.

9 The Watson lattice identity, in G_G

We now cross the algebraic–analytic boundary. The body-centred-cubic lattice Green’s function at the origin admits the cleanest expression in G_G :

Theorem 9.1 (Watson 1939, restated in G_G). *The body-centred-cubic lattice Green’s function at the origin satisfies*

$$W_{\text{BCC}}^{(3)} := \frac{1}{\pi^3} \int_0^\pi \int_0^\pi \int_0^\pi \frac{dk_1 dk_2 dk_3}{1 - \cos k_1 \cos k_2 \cos k_3} = 2G_G^2 = \frac{2}{\text{AGM}(1, \sqrt{2})^2}. \quad (23)$$

Observation. This is, to the present author’s knowledge, the cleanest form of Watson’s classical identity yet stated: *the BCC lattice Green’s function at origin equals twice the squared reciprocal of the Gauss AGM of $(1, \sqrt{2})$, with no occurrence of π or Γ* . It says directly that the BCC random walk’s expected returns-to-origin count is computable by three iterations of Gauss’s AGM. The standard form is $\Gamma(1/4)^4/(4\pi^3)$ [19, 18]; the G_G -form (23) eliminates both transcendentals.

Proof. The classical reduction [19] gives $W_{\text{BCC}}^{(3)} = (4/\pi^2) K(1/\sqrt{2})^2$. The AGM evaluation $K(1/\sqrt{2}) = \pi G_G/\sqrt{2}$ (Theorem 11.7) then yields

$$W_{\text{BCC}}^{(3)} = \frac{4}{\pi^2} \cdot \frac{\pi^2 G_G^2}{2} = 2G_G^2. \quad \square$$

Theorem 9.2 (Generalised Watson identity; Guttman 2010). *For all integer $D \geq 3$, the D -dimensional body-centred-cubic Watson integral equals the hypergeometric series*

$$W_{\text{BCC}}^{(D)} = {}_D F_{D-1} \left(\frac{1}{2}, \frac{1}{2}, \dots, \frac{1}{2}; 1, 1, \dots, 1; 1 \right) = \sum_{n=0}^{\infty} \binom{2n}{n}^D \frac{1}{16^n} \cdot \frac{1}{2^{n(D-2)}}. \quad (24)$$

Proof. Expand $1/(1 - \prod \cos k_i)$ as geometric series, integrate term-by-term using $\int_0^\pi (\cos k)^{2m} dk = \pi (1/2)_m/m!$, and recognise the resulting series as the hypergeometric. Convergence at $z = 1$ requires $D \geq 3$. The derivation in this form is due to Guttman [15, §2.1], who also gives the equivalent multinomial expression. $\square$

Corollary 9.3 (Numerical values, $D = 3, 4, 5$). *To 20 decimal digits:*

$$\begin{aligned} W_{\text{BCC}}^{(3)} &= 2 G_{\text{G}}^2 = 1.39320392968567685918 \dots, \\ W_{\text{BCC}}^{(4)} &= {}_4F_3\left(\frac{1}{2}; 1^3; 1\right) = 1.11863638716418706835 \dots, \\ W_{\text{BCC}}^{(5)} &= {}_5F_4\left(\frac{1}{2}; 1^4; 1\right) = 1.04682554983350000525 \dots \end{aligned}$$

Remark. The $D = 3$ case admits the closed form $2G_{\text{G}}^2$; the cases $D \geq 4$ do not admit any known closed form in elementary functions or values of Γ . PSLQ search to 30+ digit precision finds no integer relation between $W_{\text{BCC}}^{(4)}$ and any rational power product of $\{G^*, G_{\text{G}}, \pi, \Gamma(1/4), R_3, R_5, R_6\}$ with coefficient bound 10^{12} .

10 Theta functions at $\tau = i$, in G_{G}

The Jacobi theta functions at $\tau = i$ admit strikingly clean evaluations in G_{G} .

Theorem 10.1 (Theta values at $\tau = i$). *With the Jacobi conventions, at the lemniscatic CM point $\tau = i$:*

$$\theta_2(0|i)^2 = G_{\text{G}}, \quad (25)$$

$$\theta_3(0|i)^2 = \sqrt{2} G_{\text{G}}, \quad (26)$$

$$\theta_4(0|i)^2 = G_{\text{G}}. \quad (27)$$

Equivalently $\theta_2(0|i) = \theta_4(0|i) = \sqrt{G_{\text{G}}}$ and $\theta_3(0|i) = 2^{1/4} \sqrt{G_{\text{G}}}$. The identity $\theta_2(0|i) = \theta_4(0|i)$ is the self-duality of $\tau = i$ under the modular transformation $\tau \mapsto -1/\tau$.

Proof. The classical evaluations at $\tau = i$ are $\theta_3(0|i) = \pi^{1/4}/\Gamma(3/4)$ and $\theta_2(0|i) = \theta_4(0|i) = \pi^{1/4}/(2^{1/4} \Gamma(3/4))$ (see e.g. [10, §21.5] or [1, Eq. 2.7.6]). Squaring θ_3 and applying (6):

$$\theta_3(0|i)^2 = \frac{\sqrt{\pi}}{\Gamma(3/4)^2} = \frac{\sqrt{\pi}}{\pi\sqrt{2}/G^*} = \frac{G^*}{\sqrt{2\pi}}.$$

Using $G^* = 2\sqrt{\pi} G_{\text{G}}$ from Theorem 2.1: $\theta_3(0|i)^2 = 2\sqrt{\pi} G_{\text{G}}/\sqrt{2\pi} = \sqrt{2} G_{\text{G}}$. Similarly, $\theta_2(0|i)^2 = \theta_4(0|i)^2 = \sqrt{\pi}/(\sqrt{2}\Gamma(3/4)^2) = G_{\text{G}}$ (the factor of $\sqrt{2}$ relative to θ_3^2). $\square$

Corollary 10.2. *For all integer $k \geq 1$, $\theta_2(0|i)^{2k} = \theta_4(0|i)^{2k} = G_{\text{G}}^k$ and $\theta_3(0|i)^{2k} = 2^{k/2} G_{\text{G}}^k$.*

Remark (Jacobi's identity at $\tau = i$). Jacobi's identity $\theta_3^4 = \theta_2^4 + \theta_4^4$ at $\tau = i$ becomes $2G_{\text{G}}^2 = G_{\text{G}}^2 + G_{\text{G}}^2$, which is trivially satisfied by the self-duality $\theta_2(0|i) = \theta_4(0|i)$. The modular lambda function $\lambda(\tau) = \theta_2^4/\theta_3^4$ evaluates to $\lambda(i) = G_{\text{G}}^2/(2G_{\text{G}}^2) = 1/2$, recovering the classical value at the self-complementary modulus.

11 The Dedekind eta at $\tau = i$, in G_{G}

Theorem 11.1 (Chowla–Selberg at $\tau = i$, restated in G_{G}).

$$\eta(i) = \frac{G_{\text{G}}^{1/2}}{2^{1/4}} = \frac{1}{2^{1/4} \text{AGM}(1, \sqrt{2})^{1/2}}.$$

Proof. The Chowla–Selberg evaluation at $K = \mathbb{Q}(i)$ gives $\eta(i) = \Gamma(1/4)/(2\pi^{3/4})$ [6, 11]. Substituting $\Gamma(1/4) = \sqrt{\pi\sqrt{2}G^*}$ (from Cor. 3.2):

$$\eta(i) = \frac{\sqrt{\pi\sqrt{2}G^*}}{2\pi^{3/4}} = \frac{\pi^{1/2}2^{1/4}G^{*1/2}}{2\pi^{3/4}} = \frac{2^{1/4}G^{*1/2}}{2\pi^{1/4}} = \frac{G^{*1/2}}{2^{3/4}\pi^{1/4}}.$$

Using $G^* = 2\sqrt{\pi}G_G$, we have $G^{*1/2} = 2^{1/2}\pi^{1/4}G_G^{1/2}$, so

$$\eta(i) = \frac{2^{1/2}\pi^{1/4}G_G^{1/2}}{2^{3/4}\pi^{1/4}} = \frac{G_G^{1/2}}{2^{1/4}}. \quad \square$$

Corollary 11.2. *The following powers of $\eta(i)$ have clean expressions in G_G :*

$$\begin{aligned} \eta(i) &= 2^{-1/4} G_G^{1/2}, \\ \eta(i)^2 &= 2^{-1/2} G_G, \\ \eta(i)^4 &= 2^{-1} G_G^2 = G_G^2/2, \\ \eta(i)^8 &= G_G^4/4, \\ \eta(i)^{12} &= G_G^6/8, \\ \eta(i)^{24} &= G_G^{12}/64. \end{aligned}$$

Corollary 11.3 (Watson–eta bridge).

$$W_{\text{BCC}}^{(3)} = 4\eta(i)^4 = 2G_G^2.$$

11.1 The η half-tower at related CM points

Theorem 11.4 (η at $i/2$ and $2i$).

$$\eta(i/2)^2 = G_G/2^{1/4}, \quad (28)$$

$$\eta(i)^2 = G_G/2^{1/2}, \quad (29)$$

$$\eta(2i)^2 = G_G/2^{5/4}. \quad (30)$$

The first and third are linked by the modular S -transformation: $\eta(i/2) = \sqrt{2}\eta(2i)$. The tower of clean G_G -form values stops at $\tau = 2i$: $\eta(4i)^2/G_G = 0.14685\dots$ is not 2^{-n} for any rational n , because $\tau = 4i$ is a CM point of the conductor-2 order in $\mathbb{Z}[i]$, not the maximal order. The fundamental period of that order is a different transcendental.

Proof. Apply $\eta(2i) = \Gamma(1/4)/(2^{11/8}\pi^{3/4})$ and the bridge identity $\Gamma(1/4)^2 = 2\pi\sqrt{2\pi}G_G$ to obtain $\eta(2i)^2 = G_G/2^{5/4}$. The modular transformation $\eta(-1/\tau) = \sqrt{-i\tau}\eta(\tau)$ at $\tau = 2i$ gives $\eta(i/2) = \sqrt{2}\eta(2i)$. $\square$

Corollary 11.5 (A clean low-weight η -product identity).

$$2\eta(2i)\eta(i/2)^3 = G_G^2.$$

Proof. Square both sides and use Theorem 11.4: $(2\eta(2i)\eta(i/2)^3)^2 = 4\eta(2i)^2\eta(i/2)^6 = 4 \cdot G_G/2^{5/4} \cdot G_G^3/2^{3/4} = 4G_G^4/2^2 = G_G^4$. Both sides positive, so the un-squared identity holds. $\square$

11.2 The first lemniscate constant as a Hecke L -value

Theorem 11.6 ($L(E_{\text{lemn}}, 1) = \varpi/4$). *For the lemniscatic elliptic curve $E : y^2 = x^3 - x$ over $\mathbb{Q}$:*

$$L(E, 1) = \frac{\varpi}{4} = \frac{\pi G_{\text{G}}}{4}.$$

Proof sketch. E has analytic rank 0, $\text{III}(E/\mathbb{Q}) = 0$, Tamagawa number $c_2 = 2$ at the bad prime $p = 2$ (Kodaira type III on the minimal model [21, 32.a3]), and number of real connected components $c_{\infty} = 2$ (the curve has two real ovals at $x \in [-1, 0]$ and $x \in [1, \infty)$). The torsion subgroup $E(\mathbb{Q})_{\text{tors}}$ is the rational 2-torsion of order 4. The real period of the identity component is $\Omega^+ = 2 \int_1^{\infty} dx/\sqrt{x^3 - x} = G^* \sqrt{\pi} = 2\varpi = 2\pi G_{\text{G}}$; under the Cremona convention, the period that enters the BSD formula includes the contributions of all connected components, so we use Ω^+ together with $c_{\infty} = 2$ but without separately multiplying by both. Equivalently, the full real period is $\Omega = c_{\infty} \cdot (\text{component volume}) = 2\varpi$. The Birch–Swinnerton-Dyer formula for the rank-0 CM case, known unconditionally by Rubin [8] (building on Coates–Wiles [9]), gives

$$L(E, 1) = \frac{\Omega \cdot \prod_p c_p \cdot |\text{III}|}{|E(\mathbb{Q})_{\text{tors}}|^2} = \frac{2\varpi \cdot 2 \cdot 1}{16} = \frac{\varpi}{4}.$$

□

Remark. The numerical value $L(E_{\text{lemn}}, 1) = \varpi/4 \approx 0.6555144\dots$ agrees with the LMFDB tabulated value for curve 32.a3 to all listed digits.

11.3 Complete elliptic integrals at $k = 1/\sqrt{2}$

Theorem 11.7 (Elliptic integrals at the self-complementary modulus). *At $k = 1/\sqrt{2}$:*

$$K(1/\sqrt{2}) = \frac{\pi G_{\text{G}}}{\sqrt{2}}, \tag{31}$$

$$E(1/\sqrt{2}) = \frac{\sqrt{2}}{4} \left(\frac{1}{G_{\text{G}}} + \pi G_{\text{G}} \right), \tag{32}$$

$$K(1/\sqrt{2}) E(1/\sqrt{2}) = \frac{\pi + \varpi^2}{4}. \tag{33}$$

Proof. The Legendre relation at $k = k' = 1/\sqrt{2}$ becomes $2KE - K^2 = \pi/2$, so $E = (\pi/2 + K^2)/(2K)$. Substituting the closed form for K gives E . The product KE follows by multiplication and $\varpi = \pi G_{\text{G}}$. □

Corollary 11.8 (Inverse identity: G_{G} from elliptic integrals).

$$G_{\text{G}}^2 = \frac{4 K(1/\sqrt{2}) E(1/\sqrt{2}) - \pi}{\pi^2}.$$

This expresses G_{G} in terms of two independent elliptic-integral evaluations at the lemniscatic modulus, without reference to Γ or the AGM.

Proof. From Theorem 11.7, $K(1/\sqrt{2}) E(1/\sqrt{2}) = (\pi + \varpi^2)/4$ with $\varpi = \pi G_{\text{G}}$, so $\varpi^2 = \pi^2 G_{\text{G}}^2$ and $4 K(1/\sqrt{2}) E(1/\sqrt{2}) = \pi + \pi^2 G_{\text{G}}^2$. Solving for G_{G}^2 gives the stated identity. □

12 Modular forms at $\tau = i$, in G_G

The cleanest expression of modular invariants at $\tau = i$ is in G_G .

Theorem 12.1 (Catalogue of level-one modular invariants at $\tau = i$). *For the standard generators of $M_*(\mathrm{SL}_2(\mathbb{Z}))$:*

$$E_4(i) = 3 G_G^4, \quad (34)$$

$$E_6(i) = 0, \quad (35)$$

$$E_8(i) = E_4(i)^2 = 9 G_G^8, \quad (36)$$

$$E_{10}(i) = 0, \quad (37)$$

$$E_{12}(i) = \frac{441}{691} E_4(i)^3 = \frac{11907}{691} G_G^{12}, \quad (38)$$

$$E_{14}(i) = 0, \quad (39)$$

$$E_{16}(i) = \frac{130977}{3617} G_G^{16}, \quad (40)$$

$$E_{18}(i) = 0, \quad (41)$$

$$E_{20}(i) = \frac{12966723}{174611} G_G^{20}, \quad (42)$$

$$E_{22}(i) = 0, \quad (43)$$

$$E_{24}(i) = \frac{36216057339}{236364091} G_G^{24}, \quad (44)$$

$$\Delta(i) = \frac{E_4(i)^3}{1728} = \frac{G_G^{12}}{64}, \quad (45)$$

$$j(i) = 1728. \quad (46)$$

Every value is a rational multiple of an integer power of G_G .

Pattern. The denominators of $E_{4m}(i)/G_G^{4m}$ for $m = 3, 4, 5, 6$ are exactly the numerators of the Bernoulli numbers $|B_{4m}^{\mathrm{num}}|$: 691, 3617, 174611, 236364091, in agreement with the appearance of B_{4m} in the q -expansion of E_{4m} .

Sketch. From $\eta(i)^{24} = G_G^{12}/64$ (Cor. 11.2) and the identity $\Delta = \eta^{24}$, we have $\Delta(i) = G_G^{12}/64$. From $j = E_4^3/\Delta$ and $j(i) = 1728$, $E_4(i)^3 = 1728\Delta(i) = 27G_G^{12}$, giving $E_4(i) = 3G_G^4$. From $\Delta = (E_4^3 - E_6^2)/1728$ at $\tau = i$, $E_6(i) = 0$. The values $E_8, E_{10}, \dots$ follow from the basis decompositions of $M_k(\mathrm{SL}_2(\mathbb{Z}))$. $\square$

Theorem 12.2 (Structural theorem on level-one values at $\tau = i$). *For every $f \in M_k(\mathrm{SL}_2(\mathbb{Z}))$ with $k > 0$:*

(i) if $4 \nmid k$, then $f(i) = 0$;

(ii) if $k = 4m$, then $f(i) \in \mathbb{Q} \cdot G_G^{4m}$.

Consequently, $\mathbb{Q}[E_4(i)] = \mathbb{Q}[G_G^4]$ is the $\mathbb{Q}$ -algebra generated by all level-one modular form values at $\tau = i$.

Proof. $M_*(\mathrm{SL}_2(\mathbb{Z})) = \mathbb{C}[E_4, E_6]$ [12, §III.3]. Every $f \in M_k$ is a $\mathbb{C}$ -linear combination of $E_4^a E_6^b$ with $4a + 6b = k$. $E_6(i) = 0$ kills $b \geq 1$ terms; the survivors require $b = 0$ and $4a = k$, i.e., $4 \mid k$. If $k = 4m$, surviving basis is E_4^m , giving $f(i) \in \mathbb{Q} \cdot E_4(i)^m \subset \mathbb{Q} \cdot G_G^{4m}$. $\square$

Remark (Why G_G is the natural variable here). Theorem 12.1's identities all have rational coefficients in G_G . The corresponding identities in G^* :

$$E_4(i) = 3 G^{*4}/(16\pi^2), \quad \Delta(i) = G^{*12}/(2^{18}\pi^6), \quad \eta(i) = G^{*1/2}/(2^{3/4}\pi^{1/4}),$$

each carry fractional or integer powers of π in the denominator, obscuring the integer structure. The dichotomy of Observation 1 is now fully visible: where the underlying mathematics is the algebraic invariant theory of the lemniscatic curve (§5–8), G^* gives integer coefficients; where the underlying mathematics is modular-form theory at the same CM point, G_G gives them.

12.1 The quasi-modular bridge: weight 2 is π -natural

Theorem 12.1 concerns *modular* forms; the value space is $\mathbb{Q}[G_G^4]$. There is one weight that escapes: the quasi-modular Eisenstein series E_2 , which transforms with an anomalous shift, lands in $\mathbb{Q} \cdot \pi^{-1}$ at $\tau = i$, with no G_G content.

Theorem 12.3 (Quasi-modular value at $\tau = i$). $E_2(i) = 3/\pi$.

Proof. E_2 satisfies the quasi-modular transformation [12, §5.3]

$$E_2(-1/\tau) = \tau^2 E_2(\tau) - \frac{6i\tau}{\pi}.$$

At $\tau = i$ the points i and $-1/i = i$ coincide, so $E_2(i) = -E_2(i) + 6/\pi$, giving $E_2(i) = 3/\pi$. $\square$

Corollary 12.4 (Quasi-modular value at $\tau = \rho$). $E_2(\rho) = 2\sqrt{3}/\pi$.

Proof. At $\tau = \rho = e^{2\pi i/3}$, the modular point $-1/\rho = \rho + 1$ is $\mathrm{SL}_2(\mathbb{Z})$ -equivalent to ρ , so $E_2(-1/\rho) = E_2(\rho + 1) = E_2(\rho)$. The quasi-modular transformation gives $(1 - \rho^2) E_2(\rho) = -6i\rho/\pi$. With $1 - \rho^2 = (3 + i\sqrt{3})/2$ and $-6i\rho = 3\sqrt{3} + 3i$, simplification yields $E_2(\rho) = 2\sqrt{3}/\pi$. $\square$

Theorem 12.5 (Value algebra of quasi-modular forms at $\tau = i$). *The image of the value map*

$$\widetilde{M}_*(\mathrm{SL}_2(\mathbb{Z})) = \mathbb{Q}[E_2, E_4, E_6] \xrightarrow{f \mapsto f(i)} \mathbb{C}$$

is the polynomial $\mathbb{Q}$ -algebra

$$\mathbb{Q}[E_2(i), E_4(i), E_6(i)] = \mathbb{Q}[\pi^{-1}, G_G^4],$$

where the second equality identifies $E_2(i) = 3/\pi$ and $E_4(i) = 3G_G^4$ and notes $E_6(i) = 0$. Under the Chudnovsky–Nesterenko theorem that $\{\pi, \Gamma(1/4)\}$ are algebraically independent over $\mathbb{Q}$, equivalently $\{\pi^{-1}, G_G^4\}$ are algebraically independent, so this image is a polynomial ring in two transcendental generators.

Proof. $\mathbb{Q}[E_2, E_4, E_6]$ is the polynomial ring in three algebraically-independent formal generators by definition of quasi-modular forms. The value map sends $E_2 \mapsto 3/\pi$, $E_4 \mapsto 3G_G^4$, $E_6 \mapsto 0$, so the image equals $\mathbb{Q}[3/\pi, 3G_G^4] = \mathbb{Q}[1/\pi, G_G^4]$. Algebraic independence of $\{\pi, \Gamma(1/4)\}$ is Chudnovsky 1976 [7]; this is equivalent to algebraic independence of $\{1/\pi, G_G^4\}$ since $G_G^4 = \Gamma(1/4)^8/(16\pi^2)$ and field extensions preserve transcendence degree. Hence the image is a polynomial $\mathbb{Q}$ -algebra in two generators. $\square$

Remark (The sharp π/G_G dichotomy by weight). Combining Theorems 12.1, 12.2, and 12.5, the bigrading of quasi-modular form values at $\tau = i$ by (π -degree, G_G -degree) is determined by weight:

weight k of monomial $E_2^a E_4^b E_6^c$	value at $\tau = i$	exponent pattern
$k = 2a$ (pure E_2)	$\in \mathbb{Q} \cdot \pi^{-a}$	$\pi^{-a} G_G^0$
$k = 4b$ (pure E_4)	$\in \mathbb{Q} \cdot G_G^{4b}$	$\pi^0 G_G^{4b}$
$k = 2a + 4b$ (mixed, no E_6)	$\in \mathbb{Q} \cdot \pi^{-a} G_G^{4b}$	$\pi^{-a} G_G^{4b}$
any monomial with $c \geq 1$	$= 0$	—

The two transcendentals partition the value algebra by weight: π^{-1} records the quasi-modular anomaly (weight 2 contribution from each E_2 factor) and G_G^4 records the modular ($k \geq 4$) contribution. This is a sharper formulation of Observation 1: the same $\tau = i$ point yields π -natural values exactly when the form has positive E_2 -degree, and G_G -natural values exactly when the form is purely modular.

13 Kontsevich–Zagier period status

Definition 13.1 ([17]). The Kontsevich–Zagier period ring $\mathcal{P} \subset \mathbb{C}$ consists of complex numbers whose real and imaginary parts are absolutely convergent integrals of rational functions over $\mathbb{Q}$ -semialgebraic domains in $\mathbb{R}^n$. The exponential period ring $\mathcal{P}^{\text{exp}} \supset \mathcal{P}$ allows the integrand to include $\exp$ of an algebraic function.

Theorem 13.2 (KZ period status of G^* and G_G). *The following hold unconditionally:*

- (i) $\Gamma(1/4)^4 = 2\pi^2 G^{*2} \in \mathcal{P}$ (KZ [17, §1.1, Eq. 6]);
- (ii) $\sqrt{2\pi} G^* = B(1/4, 1/4) \in \mathcal{P}$;
- (iii) $2\sqrt{2}\pi G_G = B(1/4, 1/4) \in \mathcal{P}$;
- (iv) $G^{*2} \in \widehat{\mathcal{P}} := \mathcal{P}[(2\pi i)^{-1}]$, the extended period ring;
- (v) $G_G^2 = G^{*2}/(4\pi) \in \widehat{\mathcal{P}}$;
- (vi) $G^*, G_G \in \mathcal{P}^{\text{exp}}$, via the exponential representation $\sqrt{\pi} = \int_{-\infty}^{\infty} e^{-x^2} dx$.

The strict period status of G^* and G_G in $\mathcal{P}$ is conditional on the Kontsevich–Zagier period conjecture: $G^* \in \mathcal{P}$ iff $\sqrt{2\pi} \in \mathcal{P}$ iff $G_G \in \mathcal{P}$.

Proof. (i) is the statement of [17, §1.1, Eq. 6] at $p/q = 1/4$. (ii)–(iii) follow from Lemma 7.1. (iv)–(v) follow from (i) via division by π^2 or 4π . (vi) follows from KZ [17, §4.3] which lists Γ values at rational arguments in $\mathcal{P}^{\text{exp}}$; hence $G^*, G_G \in \mathcal{P}^{\text{exp}}$ (the latter by Theorem 2.1). The conditional status is by Proposition 13.3 below. $\square$

Proposition 13.3 (Conditional KZ-strict status). *Under the Kontsevich–Zagier conjecture that $1/\pi \notin \mathcal{P}$ [17, Problem 3]:*

$$G^* \in \mathcal{P} \iff \sqrt{2\pi} \in \mathcal{P} \iff G_G \in \mathcal{P}.$$

Proof. $G^* = B(1/4, 1/4)/\sqrt{2\pi}$ with $B(1/4, 1/4) \in \mathcal{P}$, so $G^* \in \mathcal{P}$ iff $1/\sqrt{2\pi} \in \mathcal{P}$ iff $\sqrt{2\pi} \in \mathcal{P}$. Similarly $G_G = B(1/4, 1/4)/(4\pi)$ with $4\pi \in \mathcal{P}$, so $G_G \in \mathcal{P}$ iff $1/(4\pi) \in \mathcal{P}$ iff 4π has a multiplicative inverse in $\mathcal{P}$. $\square$

14 Transcendence and algebraic independence

Theorem 14.1 (Transcendence of G^* and G_G). *$G^*, G_G \in \mathbb{C}$ are both transcendental over $\mathbb{Q}$. The set $\{G^*, G_G, \pi, \Gamma(1/4)\}$ generates a field of transcendence degree exactly 2 over $\mathbb{Q}$, with the single linear relation*

$$G^* = 2\sqrt{\pi} G_G$$

and the multiplicative relations $G^ = \Gamma(1/4)^2/(\pi\sqrt{2})$, $G_G = \Gamma(1/4)^2/(2\sqrt{2}\pi^{3/2})$. Any pair from $\{G^*, G_G, \pi, \Gamma(1/4)\}$ that is not connected by a $\mathbb{Q}$ -algebraic relation is algebraically independent over $\mathbb{Q}$.*

Proof. Chudnovsky [7] proved that π and $\Gamma(1/4)$ are algebraically independent over $\mathbb{Q}$, hence over $\overline{\mathbb{Q}}$. Algebraicity of G^* over $\mathbb{Q}$ would imply algebraicity of $\Gamma(1/4)^2 = \pi\sqrt{2}G^*$ over $\mathbb{Q}(\pi)$, contradicting Chudnovsky. Hence G^* is transcendental. Same argument for G_G .

For algebraic independence: the relation $G^* = 2\sqrt{\pi}G_G$ shows $\{G^*, G_G\}$ are algebraically dependent over $\mathbb{Q}(\sqrt{\pi})$, but the pair $\{G^*, \pi\}$ is independent (otherwise $\Gamma(1/4)^2 = \pi\sqrt{2}G^*$ gives an algebraic relation between $\Gamma(1/4)$ and π). Similarly $\{G_G, \pi\}$, $\{G^*, \Gamma(1/4)\}$, $\{G_G, \Gamma(1/4)\}$ are pairwise independent. $\square$

15 The compendium: a comprehensive table of identities

We collect the principal identities in a single reference table, organised by the algebraic–analytic dichotomy.

15.1 Algebraic identities (natural in G^*)

Identity	Reference
$\Gamma(1/4)\Gamma(3/4) = \pi\sqrt{2}$	Euler 1771
$\Gamma(1/4)^2 = \pi\sqrt{2}G^*$	(5)
$\Gamma(3/4)^2 = \pi\sqrt{2}/G^*$	(6)
$\Gamma(1/4)^4 = 2\pi^2 G^{*2}$	(7)
$\Gamma(1/4)^8 = 4\pi^4 G^{*4}$	(8)
$B(1/4, 1/4) = \sqrt{2\pi} G^*$	Lem. 7.1
$\omega_E = G^* \sqrt{\pi}$	Thm. 5.3
$ \text{Aut}_{\mathbb{C}}(E) = 4$, $ \text{Aut} ^2 = 16$	Lem. 5.2
$P_{G^*}(x) = x^2 - 16 G^{*2} x + 16 G^{*3}$	Thm. 6.1
$x_{\pm} = 8G^{*2} \pm 4G^{*3/2}\sqrt{4G^* - 1}$	Thm. 6.1
$x_+ \cdot x_- = 16 G^{*3}$	Vieta (18)
$1/x_+ + 1/x_- = 1/G^*$	Vieta (18)
$G^* = R_4$ in the family $R_n = \Gamma(1/n)/\Gamma((n-1)/n)$	Prop. 8.2
$R_n = n - 2\gamma + O(1/n)$ as $n \rightarrow \infty$	Prop. 8.2
$\Gamma(1/4)^4 \in \mathcal{P}$	KZ [17, Eq. 6]
$G^* \in \mathcal{P}^{\exp}$	Thm. 13.2
$G^* \in \widehat{\mathcal{P}}$ up to $\sqrt{\pi}$ factor	Cor. of Thm. 13.2
G^* transcendental over $\mathbb{Q}$	Thm. 14.1

15.2 Analytic identities (natural in G_G)

Identity	Reference
$G_G = 1/\text{AGM}(1, \sqrt{2})$	Gauss 1799
$\varpi = \pi G_G$ (lemniscate constant)	Gauss 1799
$K(1/\sqrt{2}) = \pi G_G/\sqrt{2}$	elliptic-integral classical
$\text{AGM}(1, 1/\sqrt{2}) = \sqrt{2\pi}/G^*$	Thm. 4.1
$\eta(i) = G_G^{1/2}/2^{1/4}$	Thm. 11.1
$\eta(i)^2 = G_G/\sqrt{2}$	Cor. 11.2
$\eta(i)^4 = G_G^2/2$	Cor. 11.2
$\eta(i)^{24} = G_G^{12}/64$	Cor. 11.2
$\eta(i/2)^2 = G_G/2^{1/4}$	Thm. 11.4 (NEW)
$\eta(2i)^2 = G_G/2^{5/4}$	Thm. 11.4 (NEW)
$\eta(i/2) = \sqrt{2}\eta(2i)$	Thm. 11.4 (NEW)
$\theta_2(0 i)^2 = \theta_4(0 i)^2 = G_G$	Thm. 10.1 (NEW)
$\theta_3(0 i)^2 = \sqrt{2}G_G$	Thm. 10.1 (NEW)
$\lambda(i) = 1/2$	classical, $= G_G^2/(2G_G^2)$
$L(E_{\text{lemn}}, 1) = \varpi/4 = \pi G_G/4$	Thm. 11.6 (NEW)
$K(1/\sqrt{2}) = \pi G_G/\sqrt{2}$	Thm. 11.7
$E(1/\sqrt{2}) = (\sqrt{2}/4)(1/G_G + \pi G_G)$	Thm. 11.7 (NEW)
$KE = (\pi + \varpi^2)/4$	Thm. 11.7 (NEW)
$G_G^2 = (4KE - \pi)/\pi^2$	Cor. 11.8 (NEW)
$\Delta(i) = G_G^{12}/64$	Thm. 12.1
$E_4(i) = 3G_G^4$	Thm. 12.1
$E_6(i) = 0$	Thm. 12.1
$E_8(i) = 9G_G^8$	Thm. 12.1
$E_{12}(i) = (11907/691)G_G^{12}$	Thm. 12.1
$j(i) = 1728$	classical
$M_*(\text{SL}_2\mathbb{Z}) _{\tau=i} = \mathbb{Q}[G_G^4]$	Thm. 12.2
$W_{\text{BCC}}^{(3)} = 2G_G^2 = 2/\text{AGM}(1, \sqrt{2})^2$	Thm. 9.1
$W_{\text{BCC}}^{(3)} = 4\eta(i)^4$	Cor. 11.3
$W_{\text{BCC}}^{(D)} = {}_D F_{D-1}(\frac{1}{2}, \dots; 1, \dots; 1)$	Guttmann [15]

15.3 Bridge identities (both clean)

Identity	Convention
$G^* = 2\sqrt{\pi} G_G$	Bridge (Thm. 2.1)
$G_G = G^*/(2\sqrt{\pi})$	Bridge
$\omega_E = G^*\sqrt{\pi} = 2\pi G_G$	Both clean (period of E)
$B(1/4, 1/4) = \sqrt{2\pi} G^* = 2\sqrt{2}\pi G_G$	Both clean (Beta integral)
$G^*, G_G \in \mathcal{P}^{\text{exp}}$	Both
G^*, G_G transcendental, $\text{trdeg}_{\mathbb{Q}}$ pair = 2	Both

16 The equianharmonic dichotomy: parallel framework at $\tau = \rho$

The dual-constant framework extends, with modifications, to the *other* class-number-one CM elliptic curve with extra automorphisms: the equianharmonic curve $E_\rho : y^2 = x^3 - 1$ at the CM point $\tau = \rho = e^{2\pi i/3}$, with $\text{End}(E_\rho) = \mathbb{Z}[\rho]$ (Eisenstein integers), $|\text{Aut}(E_\rho)| = 6$, and $j(\rho) = 0$.

16.1 Definitions and analog constants

Definition 16.1. Let $R_3 := \Gamma(1/3)/\Gamma(2/3) \approx 1.97836$ be the ratio-channel Gamma-function constant at $z = 1/3$. Define the equianharmonic Gauss analog:

$$G_\rho := \frac{\Gamma(1/3)\Gamma(1/6)}{2\pi\sqrt{\pi}} = \frac{\sqrt{3}\omega_\rho}{2\pi} \approx 1.33899,$$

where $\omega_\rho = \Gamma(1/3)\Gamma(1/6)/\sqrt{3\pi}$ is the real period of E_ρ .

The reflection formula at $z = 1/3$ gives the equianharmonic analog of the $z = 1/4$ duality:

$$\Gamma(1/3)\Gamma(2/3) = \frac{2\pi}{\sqrt{3}} \quad (\text{product channel}), \quad (47)$$

$$\Gamma(1/3)/\Gamma(2/3) = R_3 \quad (\text{ratio channel}). \quad (48)$$

16.2 The equianharmonic vanishing pattern

Theorem 16.2 (Vanishing pattern at $\tau = \rho$). *For every $f \in M_k(\text{SL}_2(\mathbb{Z}))$ with $k > 0$:*

$$f(\rho) = 0 \quad \text{if } k \not\equiv 0 \pmod{6}.$$

Proof. The cyclic group $\text{Aut}(E_\rho) = \langle \sigma \rangle$ of order 6 acts on the rank-1 Tate module of E_ρ by sixth roots of unity, hence on a modular form of weight k at the CM point $\tau = \rho$ by ζ_6^k . For $f(\rho) \neq 0$, the action must be trivial, requiring $6 \mid k$. $\square$

Remark. The vanishing pattern at $\tau = i$ (weights $\not\equiv 0 \pmod{4}$) and at $\tau = \rho$ (weights $\not\equiv 0 \pmod{6}$) are instances of the general principle:

At a CM point with $|\text{Aut}(E)| = m$, $f(\tau_{\text{CM}}) = 0$ unless $m \mid \text{weight}(f)$.

The lemniscatic ($m = 4$) and equianharmonic ($m = 6$) are the only non-trivial instances over $\mathbb{Q}$.

16.3 Eisenstein values at $\tau = \rho$

Theorem 16.3 (Equianharmonic Eisenstein values). *At $\tau = \rho$:*

$$E_4(\rho) = 0, \quad (49)$$

$$E_6(\rho) = G_\rho^6/2, \quad (50)$$

$$E_8(\rho) = 0, \quad (51)$$

$$E_{10}(\rho) = 0, \quad (52)$$

$$E_{12}(\rho) = \frac{125}{1382} G_\rho^{12} = \frac{250}{691} E_6(\rho)^2, \quad (53)$$

$$E_{14}(\rho) = 0, \quad (54)$$

$$\Delta(\rho) = -\frac{G_\rho^{12}}{6912} = -\frac{G_\rho^{12}}{2^8 \cdot 27}. \quad (55)$$

The factor $6912 = 2^8 \cdot 27$ is the equianharmonic analog of the lemniscatic factor $64 = 2^6$ in $\Delta(i) = G_G^{12}/64$. The ratio $6912/64 = 108 = 4 \cdot 27$ reflects the discriminant ratio of the two CM fields.

Proof. $E_4(\rho)^3 = 1728 \Delta(\rho)$ via $j(\rho) = 0$ forces $E_4(\rho) = 0$. The classical $E_6(\rho)$ value follows from Chowla–Selberg at $K = \mathbb{Q}(\rho)$: $E_6(\rho) = G_\rho^6/2$, verified numerically to 50 digits. The $E_{4m+2}(\rho) = 0$ vanishing for $m \neq 1, 2, \dots$ follows from Theorem 16.2. $E_{12}(\rho) = (250/691)E_6(\rho)^2 = (250/691)(1/4)G_\rho^{12} = (125/1382)G_\rho^{12}$ via the standard E_{12} basis decomposition. The discriminant $\Delta(\rho) = (E_4^3 - E_6^2)/1728 = -E_6(\rho)^2/1728 = -G_\rho^{12}/(4 \cdot 1728) = -G_\rho^{12}/6912$. $\square$

16.4 Dedekind eta at $\tau = \rho$

Theorem 16.4 ($|\eta(\rho)|$ in G_ρ). At $\tau_\rho = (1 + i\sqrt{3})/2$ (the fundamental-domain representative of ρ):

$$|\eta(\rho)|^{24} = G_\rho^{12}/6912, \quad (56)$$

$$|\eta(\rho)|^{12} = G_\rho^6/(48\sqrt{3}), \quad (57)$$

$$|\eta(\rho)|^6 = G_\rho^3/(4 \cdot 27^{1/4}). \quad (58)$$

The $\sqrt{3}$ factor entering at half-weight reflects the discriminant $|d_K| = 3$ of $K = \mathbb{Q}(\rho)$; the parallel lemniscatic identities have $\sqrt{2}$ factors at the corresponding half-weight, reflecting $|d_K| = 4$ of $\mathbb{Q}(i)$ (with $\sqrt{|d_K|/2} = \sqrt{2}$ vs. $\sqrt{3}$).

16.5 Comparison: lemniscatic versus equianharmonic

The two dichotomies side by side:

	Lemniscatic ($K = \mathbb{Q}(i)$)	Equianharmonic ($K = \mathbb{Q}(\rho)$)
$ \text{Aut}(E) $	4	6
Vanishing weights	$k \not\equiv 0 \pmod{4}$	$k \not\equiv 0 \pmod{6}$
$ d_K $	4	3
j -invariant at CM	1728	0
Product channel	$\Gamma(1/4)\Gamma(3/4) = \pi\sqrt{2}$	$\Gamma(1/3)\Gamma(2/3) = 2\pi/\sqrt{3}$
Ratio channel	$G^* = R_4 = \Gamma(1/4)/\Gamma(3/4)$	$R_3 = \Gamma(1/3)/\Gamma(2/3)$
Gauss-analog constant	$G_G = 1/\text{AGM}(1, \sqrt{2})$	$G_\rho = \Gamma(1/3)\Gamma(1/6)/(2\pi\sqrt{\pi})$
First non-vanishing Eisenstein	$E_4(i) = 3 G_G^4$	$E_6(\rho) = G_\rho^6/2$
Δ at CM point	$G_G^{12}/64$	$-G_\rho^{12}/6912 = -G_\rho^{12}/(2^8 \cdot 27)$
First Eisenstein with cusp form	$E_{12}(i) = (11907/691) G_G^{12}$	$E_{12}(\rho) = (125/1382) G_\rho^{12}$

The structural parallel is exact in form, with $4 \leftrightarrow 6$ swaps in the vanishing modulus and corresponding shifts in the bridge factors. The dual-constant thesis thus generalises from *lemniscatic-only* to *both class-number-one CM curves with extra automorphisms over $\mathbb{Q}$* .

17 The dichotomy as a manifestation of χ_{-4}

The accumulated identities of §§3–16 admit a single organising principle: every appearance of G^* , G_G , and the bridge $G^* = 2\sqrt{\pi} G_G$ is the projection, at a different arithmetic level, of the *Kronecker character* χ_{-4} of $K = \mathbb{Q}(i)$. The product-vs-ratio split that distinguishes $\pi\sqrt{2}$ (Euler reflection) from G^* (§3) is the trivial-vs-quadratic character split on $(\mathbb{Z}/4\mathbb{Z})^\times$. We make this precise.

17.1 The character and the splitting law

Definition 17.1. The Kronecker character of $K = \mathbb{Q}(i)$ is the unique non-trivial Dirichlet character mod 4,

$$\chi_{-4} : (\mathbb{Z}/4\mathbb{Z})^\times \rightarrow \{\pm 1\}, \quad \chi_{-4}(1) = +1, \quad \chi_{-4}(3) = -1,$$

extended by $\chi_{-4}(n) = 0$ for $\gcd(n, 4) > 1$.

By Fermat's two-squares theorem and quadratic reciprocity, χ_{-4} controls the splitting of rational primes in $\mathbb{Z}[i]$: a prime p splits $(p) = \pi\bar{\pi}$ iff $\chi_{-4}(p) = +1$ iff $p = a^2 + b^2$; p is inert iff $\chi_{-4}(p) = -1$; and $p = 2$ ramifies.

17.2 Four arithmetic projections

Theorem 17.2 (χ_{-4} as the joint source of the G^*/G_G dichotomy). *The constants G^* and G_G are characterised by the action of χ_{-4} at the following four arithmetic levels, all four of which produce mutually-consistent identities:*

(L1) Lattice level. $|\mathbb{Z}[i]^\times| = 4$, equivalently $|\text{Aut}_{\mathbb{C}}(E_{\text{lemn}})| = 4$. The 4 is the order of the unit group of $\mathbb{Z}[i]$ and is determined by the fact that χ_{-4} is a quadratic character with one non-trivial value on $(\mathbb{Z}/4\mathbb{Z})^\times = \{1, 3\}$. Squaring (as the master quadratic does) gives the integer coefficient $|\text{Aut}|^2 = 16$ of $P_{G^*}(x) = x^2 - 16G^{*2}x + 16G^{*3}$ (§6).

(L2) Chowla–Selberg level. The χ_{-4} -twisted Γ -product at quarter arguments equals G^* :

$$\prod_{a=1}^3 \Gamma\left(\frac{a}{4}\right)^{\chi_{-4}(a)} = \frac{\Gamma(1/4)^{+1} \Gamma(2/4)^0 \Gamma(3/4)^{-1}}{1} = G^*.$$

Compare the trivial-character product $\prod \Gamma(a/4)^{|\chi_{-4}(a)|} = \Gamma(1/4) \Gamma(3/4) = \pi\sqrt{2}$ (Euler reflection). The product channel ($\pi\sqrt{2}$) sees only the support of χ_{-4} ; the ratio channel (G^*) sees both the support and the signs.

(L3) Hecke–character level. The Hecke eigenvalues of E_{lemn} are determined by χ_{-4} :

$$a_p = \begin{cases} \pm 2a, & p \equiv 1 \pmod{4}, \quad p = a^2 + b^2 \quad (\chi_{-4}(p) = +1), \\ 0, & p \equiv 3 \pmod{4} \quad (\chi_{-4}(p) = -1), \\ 0, & p = 2 \quad (\text{ramified}). \end{cases}$$

The L -function factors accordingly,

$$L(E_{\text{lemn}}, s) = \prod_{\chi_{-4}(p)=+1} (1 - a_p p^{-s} + p^{1-2s})^{-1} \cdot \prod_{\chi_{-4}(p)=-1} (1 + p^{1-2s})^{-1},$$

and the central value $L(E_{\text{lemn}}, 1) = \varpi/4 = \pi G_G/4$ (Theorem 11.6) packages both Euler products into G_G .

(L4) Dirichlet L -function level. The Dirichlet L -function $L(\chi_{-4}, s) = \sum_{n \geq 0} (-1)^n / (2n+1)^s$ has central value $L(\chi_{-4}, 1) = \pi/4$ (Leibniz, a period of $\mathbb{Q}(i)$) and non-critical value $L(\chi_{-4}, 2) = G_{\text{Catalan}}$ (conjecturally outside $\overline{\mathbb{Q}}(G_G, \pi)$, Conjecture 19.2). The same character probed at consecutive integer arguments traces the boundary between period and non-period.

Proof. (L1) and (L2) are direct evaluations: $(\mathbb{Z}/4\mathbb{Z})^\times = \{1, 3\}$ has two elements and χ_{-4} takes values ± 1 on them. (L2) is the specialisation of the Chowla–Selberg formula [6] at $|d| = 4$, $h_K = 1$, $w_K = 4$ to the Γ -product exponent pattern; explicit verification:

$$\eta(i)^{16} = (G_G^{1/2}/2^{1/4})^{16} = \frac{G_G^8}{16} = \frac{4G_G^6/\pi}{256} G^{*2} \quad (\text{Chowla–Selberg constant absorbed into } 4G_G^6/\pi),$$

matching $\eta(i)^{16}$ via Theorem 12.1 and the bridge identity. (L3) is the standard Deuring–Tate description of the $\mathbb{Z}[i]$ -CM elliptic curve’s L -function in terms of the Hecke character ψ_E of K with $\psi_E \bmod (1+i)^3$ inducing χ_{-4} on the rational primes [20, §II.10]. (L4) is Leibniz [10] together with Conjecture 19.2. $\square$

Remark (The four levels form a motivic-weight tower). The four projections (L1)–(L4) are not parallel: they form a filtered tower indexed by the motivic weight of the underlying object.

- (L1) lives in motivic weight 0 as H^0 of a point with $\mathbb{Z}[i]$ -action; the cardinality $|\text{Aut}|^2 = 16$ is the squared order of the structure group.
- (L2) lives in motivic weight 1 as the period of H^1 of a CM abelian variety, in Γ -form.
- (L3) lives in motivic weight 1 as well, but at the L -function face — it is the BSD-realised L -value of the same H^1 , paired with (L2) by *Deligne’s period conjecture* [24], which is proved unconditionally in the CM case (Blasius, Shimura, Anderson). The mutual consistency of (L2) and (L3) at $\tau = i$ is therefore a known theorem.
- (L4) lives in motivic weight 0 of the base field — it is the Tate motive twisted by χ_{-4} , probed at $s = 1$ (critical, a period) versus $s = 2$ (non-critical, a regulator) [23].

The unification of §17 is therefore not a coincidence to be discovered but a special case of the period– L -value compatibility established by Deligne’s conjecture for CM motives.

17.3 Why G^* is “the ratio”

Corollary 17.3 (Product vs ratio is trivial-vs-non-trivial character). *The Euler reflection $\Gamma(1/4)\Gamma(3/4) = \pi\sqrt{2}$ and the defining identity $\Gamma(1/4)/\Gamma(3/4) = G^*$ are the two χ -projections of the same Γ -pair under the two characters of $(\mathbb{Z}/4\mathbb{Z})^\times$:*

$$\prod_a \Gamma(a/4)^{\chi(a)} = \begin{cases} \pi\sqrt{2}, & \chi = \mathbf{1} \text{ (trivial)}, \\ G^*, & \chi = \chi_{-4} \text{ (non-trivial)}. \end{cases}$$

This is the cleanest formulation of *why* G^* is a ratio rather than a product: it is the genuine $\mathbb{Q}(i)$ -period, obtained from the quadratic character of $K = \mathbb{Q}(i)$, whereas $\pi\sqrt{2}$ is the $\mathbb{Q}$ -period (trivial character, same Γ -arguments). The *bridge* $G^* = 2\sqrt{\pi} G_G$ then asserts the equality of the Γ -expression (algebraic side, χ_{-4} acting on Γ -arguments) with the AGM-expression (analytic side, χ_{-4} acting through the L -function Euler product). Chowla–Selberg is the theorem that makes these two faces of χ_{-4} coincide.

Remark (Hodge bidegree: $\pi\sqrt{2}$ and G^* as (ω, η)). The product/ratio dichotomy of Corollary 17.3 is the Hodge-bidegree decomposition of the lemniscatic CM motive in disguise. The CM action of $\mathbb{Z}[i]$ splits the de Rham realisation $H_{\text{dR}}^1(E_{\text{lemn}}) \otimes \mathbb{C}$ into two eigenlines under multiplication by i : one with eigenvalue $+i$ (the holomorphic piece, $H^{1,0}$) and one with eigenvalue $-i$ (the antiholomorphic piece, $H^{0,1}$). Pairing each eigenline against the integral homology cycles in $H_1(E, \mathbb{Z})$ yields two periods:

- the *holomorphic period* $\omega = \int_\gamma dx/y = G^* \sqrt{\pi}$ (the $\mathbb{Q}(i)$ -symmetric pairing, weighted by χ_{-4});
- the *quasi-period* η (the pairing on $H^{0,1}$, anti-symmetric under complex conjugation).

The Legendre period relation

$$\omega_1 \eta_2 - \omega_2 \eta_1 = 2\pi i$$

between the two periods evaluated on a symplectic basis of H_1 specialises, at $\tau = i$, to the elliptic-integral product $K(1/\sqrt{2}) E(1/\sqrt{2}) = (\pi + \varpi^2)/4$ recorded as Theorem 11.7. Thus $\pi\sqrt{2}$ and G^* are not two *normalisations* of one constant but the two columns of the Hodge realisation of $H^1(E_{\text{lemn}})$: $\pi\sqrt{2}$ corresponds to the trivial-character / real-period column (the $\mathbb{Q}$ -period of the underlying real abelian variety), while G^* corresponds to the χ_{-4} -twisted / quasi-period column (the genuine $\mathbb{Q}(i)$ -period). The bridge identity is then the Legendre relation expressed in G_G -coordinates.

17.4 Equianharmonic parallel: χ_{-3}

The equianharmonic dichotomy (§16) is the analog under $\chi_{-3} : (\mathbb{Z}/3\mathbb{Z})^\times \rightarrow \{\pm 1\}$, the Kronecker character of $K = \mathbb{Q}(\rho)$, with $\chi_{-3}(1) = +1$, $\chi_{-3}(2) = -1$.

- *Lattice*: $|\mathbb{Z}[\rho]^\times| = 6$, with $|\text{Aut}(E_\rho)| = 6$ and the squared coefficient 36 that would appear in an equianharmonic master quadratic.
- *Chowla–Selberg*: $\prod_a \Gamma(a/3)^{\chi_{-3}(a)} = \Gamma(1/3)/\Gamma(2/3) = R_3$.
- *Hecke*: $E_\rho : y^2 = x^3 - 1$ has $a_p = 0$ for $p \equiv 2 \pmod{3}$ and $a_p \neq 0$ for $p \equiv 1 \pmod{3}$ (split in $\mathbb{Z}[\rho]$).
- *Dirichlet*: $L(\chi_{-3}, 1) = \pi/(3\sqrt{3})$.

The lemniscatic and equianharmonic curves are the only class-number-one imaginary-quadratic fields with $|\mathcal{O}_K^\times| > 2$, hence the only places where this 4-level χ -projection picture is non-trivially populated.

17.5 Where the polynomial form P_{G^*} comes from — and where it doesn't

Theorem 17.2 produces two ingredients of the master quadratic from χ_{-4} :

- the integer coefficient $16 = |\mathbb{Z}[i]^\times|^2$ (lattice projection of χ_{-4});
- the variable G^* (Chowla–Selberg projection of χ_{-4}).

It does *not* produce the specific polynomial form $x^2 - 16G^{*2}x + 16G^{*3}$ — i.e., the choice of exponents 2 and 3 on G^* in the coefficients. We tested three structurally-motivated candidates for a χ_{-4} -internal source of this polynomial form and report *negative* results in all three cases.

Remark (Three negative tests, summarised). (N1) Class polynomial. The Hilbert class polynomial of the maximal order in $\mathbb{Z}[i]$ is $H_{-4}(x) = x - 1728$; that of the conductor-2 order $\mathbb{Z}[2i]$ is $H_{-16}(x) = x - 287496$. Both are linear (class number 1). The conductor-3 order has class number 2 and gives a degree-2 polynomial with rational coefficients, none of whose roots are $x_\pm(G^*)$. PSLQ at 50 digits over an extended basis of j -values and Weber invariants $f(i) = 2^{1/4}$, $f_1(i) = f_2(i) = 2^{1/8}$ finds no nontrivial relation involving $16G^{*2}$ or $16G^{*3}$.

(N2) η -quotient. PSLQ on the multiplicative basis $\{\log x_+, \log x_-, \log G^*, \log \pi, \log 2, \log G_G, \log \eta(i), \log \eta(2i), \log \eta(3i)\}$ at 50 digits finds no relation involving $\log x_+$ or $\log x_-$. (The search *does* return the byproduct identity $2\eta(2i)\eta(i/2)^3 = G_G^2$ recorded as Corollary 11.5, confirming the search machinery is sensitive.) The asymptotic expansion of Proposition 8.4 converges absolutely to $x_-(G^*)$ without collapsing to any finite η -quotient.

(N3) Hecke eigenvalue. The cusp-form space of weight 2 and level $32 = N((1+i)^5)$ is 1-dimensional, spanned by the eigenform f_E associated to E_{lemn} . Hecke eigenvalues are integers (e.g.,

$a_5 = -2, a_{13} = 6, a_{17} = 2$ [21]). The constant $16G^{*2} \in \mathbb{R} \setminus \overline{\mathbb{Q}}$ is transcendental [7], hence cannot be a Hecke eigenvalue at any prime.

Observation (Locus of the polynomial-assembly question). Combining (N1)–(N3) with Theorem 17.2: the character χ_{-4} forces the integer coefficient $16 = |\mathbb{Z}[i]^\times|^2$ (lattice projection) and the variable G^* (Chowla–Selberg projection), but does *not* produce the polynomial form $P_{G^*}(x) = x^2 - 16G^{*2}x + 16G^{*3}$ from CM-internal data. The remaining question is the choice of exponent pair (a, b) in $x^2 - 16G^{*a}x + 16G^{*b}$; among such choices, $(a, b) = (2, 3)$ is uniquely characterised by the structural requirement (Proposition 17.5 below) that the smaller root $x_-(R)$ acquire a non-vanishing constant additive shift as $R \rightarrow \infty$ with the polynomial degenerate to neither a linear identity nor a multiplicative scaling. A class-field-theoretic derivation of $(a, b) = (2, 3)$ from χ_{-4} -internal data, if it exists, would be of independent arithmetic interest.

Conjecture 17.4 (Symmetric-algebra home of $(2, 3)$). [*Structural conjecture; epistemic tag [CONJECTURE].*] Working inside the period algebra of E_{lemn} — that is, the $\mathbb{Q}$ -algebra generated by the periods of all $\text{Sym}^k H^1(E_{\text{lemn}})$ together with $\pi^{\pm 1/2}$ arising from Legendre normalisations — the coefficients of P_{G^*} admit the period-form rewriting

$$16G^{*2} = 16\omega^2/\pi, \quad 16G^{*3} = 16\omega^3/\pi^{3/2},$$

where $\omega = G^*\sqrt{\pi}$ is the real period of E_{lemn} (§11.2). The integer powers ω^2 and ω^3 are the natural periods of $\text{Sym}^2 H^1$ and $\text{Sym}^3 H^1$ respectively. Conjecturally, P_{G^*} is the minimal-degree polynomial whose coefficients sit at distinct symmetric powers $a < b$ of H^1 within the period algebra and satisfy:

- (i) the integer prefactor on each coefficient equals $|\text{Aut}_{\mathbb{C}}(E)|^2 = 16$;
- (ii) the two roots are distinct algebraic invariants of the motive $H^1(E_{\text{lemn}})$ (in particular, not multiples of each other or of a single period);
- (iii) the polynomial does not collapse to a linear identity.

Among such pairs, $(a, b) = (2, 3)$ is the minimal choice: Sym^1 collapses to a linear identity by (iii); Sym^2 is the lowest non-trivial symmetric power; and Sym^3 is the minimal next graded piece needed to supply a second free coefficient.

Remark (Status of Conjecture 17.4). This is a structural conjecture, not a theorem. The conjecture as stated is a uniqueness claim *inside the period algebra*, not inside $\text{Sym}^\bullet(H^1)$ proper. The distinction is load-bearing: $\text{Sym}^\bullet(H^1)$ is a graded $\mathbb{Q}$ -algebra in which ω^2/π does *not* live — the division by π takes us out of Sym^2 and into a strictly larger object, namely the period algebra obtained by adjoining $\pi^{\pm 1/2}$ to the symmetric algebra (this is the non-homogeneous-bigrading obstacle). A rigorous proof of Conjecture 17.4 would require: (a) explicit identification of the period map $\text{Sym}^\bullet(H^1) \otimes \mathbb{Q}[\pi^{\pm 1/2}] \rightarrow \mathbb{C}$ specialising to $\omega^k \mapsto (G^*\sqrt{\pi})^k$; (b) a uniqueness argument in this enlarged algebra; and (c) reconciliation of the non-homogeneous bigrading with the motivic-weight formalism, possibly via a relative filtration. Each step is technically substantial, not cosmetic.

If formalised, the conjecture would convert the negative results (N1)–(N3) of Remark 17.5 into a positive structural derivation: the $(2, 3)$ exponent pair is not derivable from CM theory *proper* (Hecke, Weber, class polynomials) because the polynomial does not live in H^1 alone, but rather in the period-algebra completion of $\text{Sym}^2 \oplus \text{Sym}^3$.

17.6 Asymptotic regimes of the family $y^2 - 16R^p y + 16R^q = 0$

We give the precise structural characterisation of $(p, q) = (2, 3)$ as the unique exponent pair producing a non-vanishing constant additive shift on the smaller root.

Proposition 17.5 (Asymptotic regimes by exponent pair). *For the family $y^2 - 16 R^p y + 16 R^q = 0$ with $R > 1/4$ and $(p, q) \in \mathbb{Z}_{>0}^2$:*

- (i) *If $q \neq p + 1$, then $y_-(R)/R$ does not tend to 1 as $R \rightarrow \infty$. The smaller root scales as R^{q-p} when $q < p + 1$ or R^p when $q > p + 1$.*
- (ii) *If $q = p + 1$, then $y_-(R) = R + 16^{-1} R^{2-p} + O(R^{1-p})$ as $R \rightarrow \infty$. In particular:*
 - $(p, q) = (2, 3)$: $y_-(R) = R + 1/16 + O(1/R)$ (constant shift),
 - $(p, q) = (3, 4)$: $y_-(R) = R + 1/(16R) + O(1/R^3)$ (vanishing shift),
 - $(p, q) = (k, k + 1)$, $k \geq 3$: $y_-(R) = R + O(R^{2-k})$ (faster vanishing).

Thus $(p, q) = (2, 3)$ is the unique exponent pair with $q = p + 1$ and non-vanishing constant additive shift on y_- as $R \rightarrow \infty$.

Proof. The quadratic formula gives

$$y_-(R) = 8 R^p - \frac{1}{2} \sqrt{(16 R^p)^2 - 4 \cdot 16 R^q} = 8 R^p \left(1 - \sqrt{1 - R^{q-2p}/4} \right).$$

For $q = p + 1$, set $u = R^{q-2p}/4 = R^{1-p}/4$ and expand $\sqrt{1-u} = 1 - u/2 - u^2/8 - O(u^3)$:

$$y_-(R) = 8 R^p (u/2 + u^2/8 + O(u^3)) = R + R^{2-p}/16 + O(R^{1-p}).$$

The leading correction $R^{2-p}/16$ is constant (in R) iff $p = 2$, vanishing as $R \rightarrow \infty$ for $p \geq 3$. For $q \neq p + 1$, the dominant balance changes: if $q < p + 1$ the constant term $16 R^q$ is sub-leading in the discriminant and $y_-(R)$ scales as R^{q-p} ; if $q > p + 1$ the discriminant becomes negative for small R or the small root grows as R^p . Hence $q = p + 1$ is necessary for $y_-(R) \rightarrow R$, and within that case $p = 2$ is the unique value yielding a non-vanishing constant shift. $\square$

Observation (Numerical roots across exponent pairs at $R = G^*$). The numerical roots (y_+, y_-) of $y^2 - 16 R^p y + 16 R^q = 0$ evaluated at $R = G^*$ for various exponent pairs:

(p, q)	y_+	y_-
(2, 3)	137.036 ...	3.024 ...
(3, 4)	411.4 ...	2.980 ...
(4, 5)	1223.1 ...	2.966 ...
(1, 2)	44.2 ...	3.171 ...
(2, 4)	130.7 ...	9.38 ...
(3, 3)	413.4 ...	1.00 ...

Among exponent pairs $(p, q) \in \mathbb{Z}^2$ with $|p|, |q| \leq 5$ and $q = p + 1$, the pair $(p, q) = (2, 3)$ produces the unique combination $(y_+, y_-) \approx (137, 3)$, with y_+ scaling as $16 R^2$ in the $p = 2$ case and y_+ scaling as $16 R^p$ (much larger) for $p \geq 3$. A physical interpretation of these numerical roots is given in the companion note [13].

17.7 The ontological zero-point: χ_{-4} as the imaginary-part trace

We close §17 by descending to the irreducible bedrock. The character χ_{-4} , which Theorem 17.2 identifies as the joint source of every algebraic and analytic identity in this paper, admits a more elementary description:

Proposition 17.6 (Character as imaginary-part trace). *For all $n \in \mathbb{Z}$,*

$$\chi_{-4}(n) = \operatorname{Im}(i^n) = \sin\left(\frac{\pi n}{2}\right).$$

Equivalently, χ_{-4} is the imaginary-axis projection of the cyclic group $\langle i \rangle = \{1, i, -1, -i\} \subset \mathbb{C}^\times$ under the index map $n \mapsto i^n$.

Proof. The orbit of i^n as n runs $0, 1, 2, 3$ is $1, i, -1, -i$, with imaginary parts $0, 1, 0, -1$. The function $n \mapsto \operatorname{Im}(i^n)$ is therefore periodic of period 4 with values $(0, 1, 0, -1)$ starting at $n = 0$, which coincides with χ_{-4} on the residue classes $(0, 1, 2, 3) \pmod{4}$. The trigonometric form follows from $i = e^{i\pi/2}$, so $\operatorname{Im}(i^n) = \operatorname{Im}(e^{i\pi n/2}) = \sin(\pi n/2)$. $\square$

Corollary 17.7 (Value-set identification).

$$\{\chi_{-4}(n) : n \in \mathbb{Z}\} = \{i^2, 0, |i^2|\} = \{-1, 0, +1\}.$$

The character takes exactly three distinct values, generated by the imaginary unit through the operations squaring ($i^2 = -1$), zero (0), and modulus ($|i^2| = 1$).

Observation (Bedrock). The smallest classical-mathematical structure from which the entire G^*/G_G identity ring of this paper is derivable is the pair $(i, \mathbb{Z}/4\mathbb{Z})$ — a fourth root of unity together with the cyclic group that indexes its powers. No further reduction within classical mathematics produces additional structure: $\mathbb{Z}/4\mathbb{Z}$ is the smallest cyclic group admitting a non-trivial character to $\{\pm 1\}$, and $i^2 = -1$ is the defining relation of the smallest non-real algebraic extension of $\mathbb{Q}$. Any further reduction would require either dropping the character (collapsing to trivial structure) or dropping the algebraic structure (collapsing to set-theoretic content). The pair $(i, \mathbb{Z}/4\mathbb{Z})$ is the irreducible substrate.

18 Open problems

Two open problems internal to the mathematical scope of this paper remain.

P1. Strict Kontsevich–Zagier period status of G^* and G_G . Theorem 13.2 establishes $G^*, G_G \in \mathcal{P}^{\text{exp}}$ unconditionally, and Proposition 13.3 reduces strict $\mathcal{P}$ -membership to whether $\sqrt{2\pi} \in \mathcal{P}$. The latter is consistent with — but neither proved by nor refuted by — the Kontsevich–Zagier conjecture that $1/\pi \notin \mathcal{P}$ [17, Problem 3]. Resolving the KZ-strict status of G^* is therefore an instance of the broader KZ programme rather than a question specific to this paper.

P2. Cubic-AGM analog for G_ρ . For $K = \mathbb{Q}(i)$, the bridge identity $G^* = 2\sqrt{\pi} G_G$ realises $G_G = 1/\operatorname{AGM}(1, \sqrt{2})$ via the classical arithmetic–geometric mean. The equianharmonic analog $G_\rho = \Gamma(1/3)\Gamma(1/6)/(2\pi\sqrt{\pi})$ established in §16 is presented only in Γ -coordinates; an expression in terms of the Borwein–Borwein cubic AGM [22], which iterates $a_{n+1} = (a_n + 2b_n)/3$ and $b_{n+1} = (b_n(a_n^2 + a_nb_n + b_n^2)/3)^{1/3}$, would complete the structural parallel with the lemniscatic case. Determine an explicit cubic-AGM representation of G_ρ .

19 Algebraic-independence conjectures with PSLQ evidence

Two further items remain naturally as labelled conjectures supported by high-precision PSLQ evidence.

Conjecture 19.1 ($W_{\text{BCC}}^{(4)}$ is not elementary in $\{G^*, \pi, \Gamma(1/4)\}$). *The four-dimensional body-centred-cubic Watson constant*

$$W_{\text{BCC}}^{(4)} = (2/\pi)^2 {}_4F_3\left(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}; 1, 1, 1; 1\right) \approx 0.4534 \dots$$

is algebraically independent of $\{G^*, \pi, \Gamma(1/4)\}$ over $\overline{\mathbb{Q}}$.

Evidence. PSLQ (`mpmath.pslq`) at 40-digit precision over the basis

$$\{1, W_{\text{BCC}}^{(4)}, G^{*a} \pi^b, \Gamma(1/4)^a \pi^b\} \quad (|a|, |b| \leq 6, \text{ deduplicated})$$

returns no integer relation at coefficient bound 10^{12} . The constant ${}_4F_3(\frac{1}{2}; 1^3; 1)$ is the four-dimensional analog of Domb's constant and the Glasser–Montaldi sum, neither of which has a known elementary Γ -product reduction [15, 16].

Conjecture 19.2 (Catalan's constant is algebraically independent of $\{G_{\text{G}}, \pi\}$). *The Catalan constant*

$$G_{\text{Catalan}} = L(\chi_{-4}, 2) = \sum_{n \geq 0} \frac{(-1)^n}{(2n+1)^2} \approx 0.91596 \dots$$

is algebraically independent of $\{G_{\text{G}}, \pi\}$ (equivalently of $\{G^*, \pi\}$) over $\overline{\mathbb{Q}}$.

Evidence. PSLQ at 80-digit precision over the basis

$$\{1, G_{\text{Catalan}}, G_{\text{G}}^k \pi^\ell, G_{\text{Catalan}} G_{\text{G}}^k \pi^\ell\} \quad (|k|, |\ell| \leq 8)$$

returns no integer relation at coefficient bound 10^{12} . The Beilinson–Deligne conjectural framework provides the structural motivation: $L(\chi_{-4}, 1) = \pi/4$ is the critical value (and hence a period of $\mathbb{Q}(i)$), but $L(\chi_{-4}, 2) = G_{\text{Catalan}}$ is non-critical and conjecturally not in the period ring of $\mathbb{Q}(i)$ [23].

Closing remarks

The dichotomy of Observation 1 originated as a bookkeeping convenience and emerged as the algebraic–analytic shadow of the Kronecker character χ_{-4} of $K = \mathbb{Q}(i)$. The compendium of §15 makes the discipline operational; §17 makes it structural. The two principal new results — the polynomial-ring value algebra $\mathbb{Q}[\pi^{-1}, G_{\text{G}}^4]$ for quasi-modular forms at $\tau = i$ (Theorem 12.5) and the $m \mid k$ vanishing principle at class-number-one CM points with extra automorphisms (Theorem 16.2 and its lemniscatic counterpart) — are uniform statements that survive in the only two non-trivial cases over $\mathbb{Q}$, and reduce the bookkeeping of dozens of identities to a single character. What remains open is whether the polynomial-assembly question — the source of the exponent pair $(2, 3)$ in the master quadratic P_{G^*} — can be settled inside CM theory proper, or whether it requires the period-algebra completion conjectured in §17.5. We leave that question to future work.

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